Behavioral Game Theory of the Two-Mechanism Bonded Commitment: Loss Aversion, Coordination, Peer Pressure, and Incentive Legibility

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Abstract

The bonded settlement primitive admits a mechanism-design analysis as a full-rationality mechanism: cooperation is the unique profile surviving iterated elimination of weakly dominated strategies, and the analysis assumes complete information, no non-pecuniary preferences, and no coordination failure under pessimistic beliefs. The full-rationality argument is correct on its own terms; this paper treats the same primitive from a behavioral-game-theory perspective and asks how the full-rationality analysis carries over when participants exhibit the documented departures from full rationality the experimental literature has catalogued over the past forty years.

The paper develops four claims. First, the primitive's reliance on weak dominance — not strict dominance — is more behaviorally robust than it might appear, because the relevant cognitive operation is a single comparison at the margin between cooperation and defection, amplified by loss aversion at the kink (not at the magnitude). Second, the Van Huyck–Battalio–Beil coordination-failure result, which appears to threaten the multi-party reading of the architecture, does not transfer because the per-player game in the bonded setting is dominance-solvable in a way the experimental weakest-link games are not. Third, the externality structure of atomic resolution produces peer pressure that is both economically real (the mechanism paper's analytical result) and behaviorally activated (the experimental literature on social-pressure responses). Fourth, the architecture's *incentive legibility* — the property that participants can verify the equilibrium argument with a single comparison rather than through iterated reasoning about counterparty rationality — makes it unusually well-suited to the cognitive constraints of real participants.

The paper closes with the empirical questions the architecture invites: experimental measurement of cooperation rates relative to controls without bonded backing, behavioral measurement of the weakest-link pressure activation across process-tree depths, and field measurement of the cognitive load the architecture imposes relative to the alternatives it is intended to displace.

Keywords: behavioral game theory, loss aversion, weakest-link coordination, peer pressure, incentive legibility, mechanism design, experimental economics, interface cognition

1 Introduction

Behavioral game theory [Camerer, 2003] is the empirical correction to mechanism design's full-rationality assumptions. Forty years of experimental work has documented systematic departures from the rational-choice baseline: loss aversion [Kahneman and Tversky, 1979, Tversky

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and Kahneman, 1992], framing effects, anchoring, hyperbolic discounting, fairness preferences [Fehr and Schmidt, 1999, Bolton and Ockenfels, 2000], social preferences, and coordination failures under pessimistic beliefs about co-players [Van Huyck et al., 1990]. A mechanism designed under the rational-choice assumption is robust to behavioral departures only if the departures preserve the mechanism's load-bearing equilibrium properties; many do not.

The mechanism-design treatment of the bonded primitive proves that cooperation is weakly dominant for both parties in the bonded game, and is the unique profile surviving iterated elimination of weakly dominated strategies. The proof assumes complete information, full rationality, no non-pecuniary preferences, and no coordination failure under pessimistic beliefs. The result holds within those assumptions. The present paper asks: how does the result carry over when the assumptions are weakened in the directions the experimental literature has identified as empirically relevant?

The four claims. The paper develops four claims about the behavioral robustness of the bonded architecture.

First, the bond's reliance on weak rather than strict dominance is structurally consequential because the strict-dominance gap is at the kink between cooperation (recovering the bond) and defection (forfeiting the bond), not at any magnitude comparison within either domain. Loss aversion amplifies the kink and is robust to the magnitude-domain diminishing sensitivity that prospect theory documents.

Second, the multi-party result is robust to the Van Huyck et al. [1990] coordination-failure pattern because the per-player game is dominance-solvable in the bonded setting. The classical experimental result requires individual non-dominance — a setting in which pessimistic beliefs about co-players make defection rational. The bonded setting forecloses this.

Third, the weakest-link subgame the mechanism paper analyzes is both economically real and behaviorally activated. The externality magnitude $P_i + 2G_i$ is large; the experimental literature on peer pressure shows that social-pressure activation occurs at much lower stakes. The architecture's peer-enforcement effect should therefore be expected to operate through both the economic-rationality channel (Paper A) and the behavioral peer-pressure channel.

Fourth, the architecture exhibits *incentive legibility*: the cognitive operation a participant must perform to assess the equilibrium is a single comparison at the margin. This is the minimal reasoning task in any strategic setting, and it is well within the documented capabilities of unsophisticated participants. The architecture's behavioral robustness derives substantially from this legibility property; it asks little of the participant.

Paper organization. Section 2 summarizes the settlement primitive in the register the present paper uses. Section 3 treats the weak-dominance argument's behavioral robustness, and works out the loss-aversion analysis at the kink. Section 4 treats the Van Huyck–Battalio–Beil coordination-failure objection and argues it does not transfer. Section 5 treats the externality structure of atomic resolution as behavioral peer pressure. Section 6 develops incentive legibility as the architectural property that makes the design unusually well-suited to behavioral participants. Section 7 treats interface cognition: what the architecture asks the participant to think about and how those thinking-tasks compare to the alternatives. Section 8 states the open empirical questions. Section 9 concludes.

2 The Settlement Primitive

The bonded primitive has a formal mechanism-design treatment, and its deployed Solidity implementation is verified through TLA$^+$, Halmos symbolic execution, Echidna property-based fuzzing, and Certora specification rules. We summarize the features the present paper uses.

Asymmetric bonding (Mechanism 1). For a transaction with payment $P > 0$ and cumulative upstream value $G \geq P$, the buyer locks $2P$ and the seller locks $2G$. The payoff matrix in the two-party game is:

	$a_B = \mathrm{C}$	$a_B = \mathrm{D}$
$a_S = \mathrm{C}$	$(-P,\ +P)$	$(-2P,\ -2G)$
$a_S = \mathrm{D}$	$(-2P,\ -2G)$	$(-2P,\ -2G)$

Cooperation weakly dominates defection for both players. Cooperation is the unique profile surviving iterated elimination of weakly dominated strategies.

Buyer dominance with atomic resolution (Mechanism 2). Only the buyer can trigger resolution; resolution settles all active orders simultaneously or not at all. The atomicity rule induces a weakest-link subgame among sellers. Each seller's defection imposes externality $\varepsilon_{k \to i} = P_i + 2G_i$ on every co-seller.

What the behavioral analysis adds. The full-rationality analysis assumes participants compute the matrix above, recognize the weak-dominance relations, and play cooperatively. The behavioral question is whether participants who do not perform this computation explicitly nonetheless converge on cooperation, and how they do so. The answer, we will argue, is that they do, through cognitive shortcuts that the mechanism's structure makes reliable.

3 Weak Dominance and Loss Aversion at the Kink

The mechanism paper's central equilibrium result rests on weak dominance: cooperation is at least as good as defection for both players, with strict inequality when the counterparty cooperates. Behavioral game theory has historically been wary of mechanisms that rely on weak dominance, because the fully-strict case is more robust to the cognitive limitations and non-pecuniary preferences participants exhibit. We argue the bonded mechanism's weak-dominance is more robust than it appears.

The strictness gap is at the kink, not the magnitude. The strict-inequality case occurs when the counterparty cooperates: cooperation yields recovery of the bond plus the net flow ($-P$ for buyer, $+P$ for seller); defection yields forfeiture of the full bond ($-2P$ for buyer, $-2G$ for seller). The gap between these outcomes is large: for the buyer, the gap is P; for the seller, the gap is $P + 2G$, which scales with chain depth via the cumulative-value accumulator. The gap is large in absolute terms (recover the bond or lose it entirely) and is large relative to the natural reference point (the bond posted at commitment).

The non-strict case occurs when the counterparty defects: both cooperation and defection yield the full bond loss because the defection prevents resolution regardless of the participant's play. This case is non-strict, but it is also the case in which the participant has no decision to make: their payoff is determined by the counterparty's defection, and their own action is moot.

Loss aversion sharpens the strict case. Prospect theory [Kahneman and Tversky, 1979, Tversky and Kahneman, 1992] establishes that participants weight losses more heavily than equivalent gains, with empirical loss-aversion coefficients typically in the range 1.5 to 2.5. The asymmetry is between the gain domain and the loss domain; within each domain, prospect theory documents diminishing sensitivity (the value function is concave over gains and convex over losses).

The architectural question is: which feature of prospect theory does the bonded mechanism's strict-dominance gap engage — the loss-aversion asymmetry, or the within-domain diminishing sensitivity? An earlier framing of this paper conflated the two. The correct analysis is that the bonded mechanism engages the loss-aversion asymmetry at the kink between cooperation and defection, while the within-domain diminishing sensitivity operates on differences within either domain (e.g., the difference between losing $2P$ now and losing $2P$ later) and does not enter the architectural analysis.

The participant's choice between cooperation and defection is a choice between a small loss (the net flow $-P$ for the buyer) and a large loss (the bond forfeiture $-2P$). On prospect theory's loss-domain value function, the disutility of a loss $x \geq 0$ is approximately $\lambda \cdot |x|^\alpha$, where $\lambda \in [1.5, 2.5]$ is the loss-aversion ratio (the asymmetry between gain and loss domains at the reference point) and $\alpha \approx 0.88$ is the loss-domain curvature parameter [Tversky and Kahneman, 1992]. The disutility of cooperation is $\lambda \cdot P^\alpha$ and the disutility of defection is $\lambda \cdot (2P)^\alpha = 2^\alpha \cdot \lambda \cdot P^\alpha$. The ratio of the disutilities is $2^\alpha \approx 2^{0.88} \approx 1.84$, which places the defection disutility roughly 84% above the cooperation disutility. The diminishing-sensitivity discount within the loss domain ($\alpha < 1$) does flatten the comparison, but not nearly enough to defeat it: $2^\alpha > 1$ for any $\alpha > 0$, so the larger loss is strictly more disutile than the smaller for any feasible curvature parameter the empirical literature has documented. Loss aversion at the kink (the λ asymmetry between domains) does no work in this comparison since both outcomes are in the loss domain; what does the work is the magnitude difference, weighted by $2^\alpha > 1$.

Behavioral robustness of weak dominance. The combined effect is that participants treat cooperation as strictly preferred to defection in the cases where the mechanism-paper analysis says cooperation strictly dominates, and treat the two as equivalent in the cases where the mechanism-paper analysis says they are equivalent (both yield bond loss because the counterparty defected). The behavioral prediction matches the rational-choice prediction at the operationally-relevant choice point.

Non-pecuniary preferences. The full-rationality analysis assumes participants care only about pecuniary payoffs. The experimental literature documents substantial non-pecuniary preferences: fairness [Fehr and Schmidt, 1999, Bolton and Ockenfels, 2000], reciprocity, spite, and intrinsic motivation. The honest scope-claim is that the bonded mechanism's equilibrium argument assumes pecuniary preferences and may admit additional behavior under non-pecuniary preferences. Three observations bound the concern.

First, fairness preferences in inequity-aversion form [Fehr and Schmidt, 1999] predict cooperation when the symmetric case applies (root commitment with $P = G$, both parties bonding $2P$). The bonded mechanism is symmetric at root and asymmetric only deeper in the process tree, where the inequity-aversion concern is the seller's higher exposure relative to the buyer's. Inequity aversion in this direction would predict a participant refusing to enter the bonded commitment in the first place rather than defecting after entering; the equilibrium-of-play result is robust to this correction at the cost of slower multi-party-process formation.

Second, reciprocity and spite operate primarily in the counterparty's-defection case where both cooperation and defection yield bond loss for the participant; the participant who is spite-motivated may prefer the defection branch as a co-counterparty-punishment gesture. The mechanism's outcome is unchanged (both branches yield bond loss); the participant's internal experience may differ.

Third, intrinsic motivation to perform (a craft-based or honor-based preference for delivering the agreed work regardless of contractual incentive) reinforces cooperation; the bonded mechanism's pecuniary incentive aligns with the intrinsic motivation rather than competing against it. This is structurally favorable for the mechanism's behavioral robustness.

4 The Van Huyck Coordination Failure Does Not Transfer

The classical experimental literature on weakest-link coordination, beginning with Van Huyck et al. [1990], documents a robust empirical finding: in groups larger than two, weakest-link coordination games converge to the Pareto-inferior all-defect equilibrium via pessimistic beliefs about co-players. The result appears to threaten the multi-party reading of the bonded architecture, since the architecture is described in the mechanism paper as inducing a weakest-link subgame among sellers.

The agent's reading of the relevant remark in the mechanism paper flagged this potential threat directly. We develop the response here in behavioral-game-theory register.

Why Van Huyck's coordination failure occurs. The Van Huyck–Battalio–Beil minimum-effort game asks each player to choose an effort level from $\{1, 2, \ldots, 7\}$. Each player's payoff is determined by the minimum effort across the group: they gain a per-player benefit from the minimum and pay a private cost for their own effort. In the typical experimental treatment, an effort of 7 across the group is Pareto-optimal, but any individual player can guarantee a non-negative payoff by choosing the minimum effort 1 regardless of others' choices. The combination produces multiple Nash equilibria (any common effort level), and the empirical finding is that groups of 14–16 players converge on the all-1 Pareto-inferior equilibrium via pessimistic beliefs about co-players' choices.

The structural feature that produces the failure is that *individual cooperation is not dominated* by individual defection. A player who chooses 7 against a co-player who chooses 1 takes a private cost without producing the joint benefit; choosing 7 is rational only conditional on others' choosing 7. Pessimistic beliefs about others' rationality therefore rationalize choosing 1, and the iterated belief-formation drives the convergence.

Why the bonded mechanism does not exhibit it. The bonded mechanism's per-seller game is structurally different. Consider the sub-game from seller S_i's perspective in the N-party process tree. Two cases.

Case 1: All other sellers cooperate, buyer resolves. Then:

$$u_{S_i}(\mathrm{C}) = +P_i,$$
$$u_{S_i}(\mathrm{D}) = -2G_i.$$

Cooperation strictly dominates defection (gap $P_i + 2G_i$).

Case 2: At least one other seller defects, or buyer defects. Then:

$$u_{S_i}(\mathrm{C}) = u_{S_i}(\mathrm{D}) = -2G_i.$$

Cooperation and defection yield the same payoff because resolution does not occur regardless of S_i's action.

The combination is weak dominance: cooperation is at least as good as defection in every co-player profile, with strict inequality in the cooperative profile. *There is no profile in which cooperation is dominated by defection.* The pessimistic-belief mechanism that produces Van Huyck-style coordination failure requires individual non-dominance — a setting in which a player who expects others to defect prefers to defect themselves. In the bonded mechanism, a player who expects others to defect is indifferent between cooperating and defecting; they are not *better off* defecting. The pessimistic-belief mechanism has nothing to operate on.

The architectural reading. The Van Huyck result documents what happens when the per-player game is non-dominance-solvable and players reason about each other's rationality under pessimistic priors. The bonded mechanism is dominance-solvable at the per-player level. The weakest-link *outcome* (one defection collapses all payoffs) is preserved; the weakest-link *strategic structure* that produces coordination failure is foreclosed. Hirshleifer's [Hirshleifer, 1983] characterization of weakest-link public-goods games as a structural-payoff form continues to apply; the experimental literature's coordination-failure transfer does not.

What this leaves open. We do not claim the bonded mechanism is fully experimentally verified. The argument above is theoretical: the structural condition for Van Huyck coordination failure is absent in the bonded mechanism. Whether participants in a bonded mechanism in fact converge on cooperation at the rates the dominance-solvability argument predicts is an experimental question Section 8 returns to. The honest claim is that the negative experimental finding from the existing weakest-link literature does not transfer; the positive experimental finding for the bonded mechanism remains to be established empirically.

5 Atomic Resolution as Peer Pressure

The mechanism paper's analysis of atomic resolution characterizes the externality structure formally: each seller's defection imposes externality $\varepsilon_{k\to i} = P_i + 2G_i$ on every other seller. The external pressure is *economic*, in the sense that it operates on each seller's expected bond exposure. The behavioral question is whether the pressure is also *social*, in the sense that participants experience it as peer pressure rather than only as expected-value arithmetic. We argue it is.

The behavioral peer-pressure literature. The experimental literature on social-pressure activation [Cialdini and Goldstein, 2004, Fehr and Gächter, 2000] finds that participants modify their behavior in response to peer monitoring even at low stakes, and that the response increases with the stakes faced by the monitor as well as by the monitored. The seminal Fehr–Gächter result on punishment in public-goods games shows that participants will pay private costs to punish defectors even when no monetary reward is available; this establishes that peer pressure activates as a behavioral phenomenon even without an expected-value rationale.

The bonded mechanism's externality structure activates a stronger form of the same phenomenon. Each seller has an *expected-value* reason to monitor co-sellers: their bond is forfeit if any co-seller defects. This is unusually high stakes: each seller faces $P_i + 2G_i$ in damages from any co-seller's defection, where G_i scales with chain depth. At chain depth $i = 5$ with equal payments p per stage, the damages are $11p$, and at $i = 10$ they are $21p$. The behavioral peer-pressure literature operates at much lower stakes; the bonded mechanism's stakes are an order of magnitude higher.

The social-substrate analogue. Joint-liability microfinance [Ghatak, 1999, Stiglitz, 1990] and the empirical literature on the Grameen Bank document a substantial reduction in reported default rates under joint-liability lending. Yunus [1999] reports a default rate of approximately 2% under the joint-liability regime, compared to industry-conventional 20% benchmarks for unsecured rural lending; Morduch [1999] contests the headline figures on accounting grounds, arguing that loan rescheduling and definition-of-default ambiguity make the self-reported 2% figure optimistic. We treat the 20% → 2% comparison as a stylized benchmark of the joint-liability literature's headline finding rather than as a peer-reviewed empirical estimate; the contemporary literature attributes whatever reduction is real to a combination of social peer-monitoring activation, social-sanction technology, and reputation effects.

The mechanism paper's reduction-from-microfinance result (Proposition 5.13 in Paper A) shows that the bonded mechanism reproduces the equilibrium peer-pressure outcome of joint-liability microfinance under strictly weaker assumptions: no repeated interaction, no local information, no exogenous punishment technology, no joint-liability contracting. The behavioral reading of the same result is that the social-pressure mechanism the microfinance literature documents is, in the bonded setting, constituted directly by the bonding rule rather than mediated through the social substrate. The bonded mechanism is therefore not merely *equivalent* to joint-liability microfinance in equilibrium; it produces the social-pressure dynamics through a structurally cleaner channel that does not require the social substrate to be in place.

Empirical extrapolation. Read as a stylized upper-bound benchmark for what joint-liability-style peer pressure has been claimed to achieve, the Grameen 20% $\rightarrow$ 2% comparison gives an indication of the order of magnitude the bonded mechanism's peer-pressure activation might support. Three structural margins favor the bonded mechanism over the Grameen setting: the externality magnitude scales with chain depth (deeper sellers face larger co-seller-defection costs); information availability is higher (every other seller's bond exposure is on chain); and the punishment technology is mechanical (the bonding rule rather than social sanction). We do not claim a specific quantitative prediction. We note that the architectural features that produced whatever effect Grameen achieves are present, in the bonded mechanism, in strengthened form, and that the Morduch-style accounting concerns about Grameen's headline figures do not apply in the same way to a mechanism whose default and resolution events are on-chain and unambiguous.

6 Incentive Legibility

The behavioral robustness of the bonded mechanism rests on a property we call *incentive legibility*: the cognitive operation a participant must perform to assess the equilibrium is a single comparison at the margin, not a multi-step iterated reasoning about counterparty rationality.

The cognitive demand of standard mechanism design. A typical mechanism-design proof requires the participant to reason about: their own preferences over outcomes; the counterparty's preferences over outcomes; their belief over the counterparty's choices; the counterparty's belief over their own choices; the equilibrium concept the mechanism rests on; and any common-knowledge assumptions the equilibrium requires. The behavioral literature documents that participants do not perform this reasoning in any reliable way — the level-k thinking literature [Stahl and Wilson, 1995, Costa-Gomes et al., 2001] finds that most participants reason at level 1 or 2 (one or two steps of iterated thinking), not the unbounded level the common-knowledge equilibrium concept presumes.

The architectural concern is that mechanisms whose equilibrium properties require unbounded iterated reasoning are not robust to the cognitive limitations of real participants. The bonded mechanism is structurally different: the equilibrium argument requires only level-1 reasoning.

The level-1 argument for cooperation. A participant in the bonded mechanism asks: "If I cooperate, what is my outcome? If I defect, what is my outcome?" The first answer is bounded by what the bonding rule produces conditional on the counterparty's action; the second answer is the bond loss unconditional on the counterparty's action. The participant compares the two and chooses the better one. This is level-1 reasoning: one step of forward-looking analysis, no iterated reasoning about the counterparty's reasoning, no common-knowledge assumption.

The argument's robustness follows: a participant who can perform level-1 reasoning — which is the floor of strategic reasoning in any documented adult population — has all the cognitive machinery the bonded mechanism requires. The mechanism does not ask the participant to compute the iterated elimination or to verify common knowledge of payoffs; it asks them to compare two outcomes at the margin. The dominance argument the mechanism paper formalizes corresponds, behaviorally, to the comparison the participant naturally performs.

The Schelling-focal-point comparison. Schelling-focal-point coordination [Schelling, 1960, Sugden, 1995] similarly works through a level-1 argument: the participant identifies the salient solution and chooses it, expecting the counterparty to do the same. The bonded mechanism's coordination is even simpler than focal-point coordination because it does not require any salience computation. The architecturally privileged choice is determined by the mechanism's payoff structure rather than by any framing- or attention-dependent salience. The participant's choice-of-cooperation is not a guess about what the counterparty will do; it is the choice that produces the better outcome conditional on either counterparty action.

Implications for the architecture's adoption. A mechanism's adoption in a population is partly a function of how well the population can verify the mechanism's equilibrium properties. A mechanism whose equilibrium argument is opaque will struggle to attract participants regardless of the argument's correctness, because participants who cannot verify the argument cannot rationally trust it. The bonded mechanism's incentive legibility is therefore not just behavioral robustness but adoption advantage: participants can verify the argument themselves with the cognitive operations they routinely perform.

7 Interface Cognition

The architecture's behavioral properties also depend on the *interface* through which participants interact with the mechanism. A clean equilibrium argument can be defeated by an interface that obscures the relevant choice or surfaces irrelevant information. We treat interface cognition briefly because it is where the architecture meets real participants.

What the architecture asks the participant to think about. The participant in a bonded commitment is asked to attend to four things at the moment of commitment: (i) the goods or services being exchanged, (ii) the price, (iii) the bond posture (which is mechanically determined by the price and the cumulative-value context), and (iv) the resolution path (which is the buyer's signature plus any clauses the parties have agreed). The mechanism paper's invariants are encoded in the structure rather than left to the participant's verification: the bonding rule produces $2P$ and $2G$ automatically, the buyer-dominance rule constrains who can resolve, and the atomic-resolution rule constrains how resolution proceeds.

The participant does not need to verify the mechanism; the mechanism enforces itself. The participant's cognitive task is limited to the substantive commercial decision (is this exchange worth this price?) plus the bond-feasibility decision (is the bond within my treasury capacity?). Both are routine commercial decisions that participants are well-equipped to perform.

Comparison with the platform-mediated alternative. A participant in a contemporary platform-mediated commerce arrangement is typically asked to attend to: the goods or services, the price, the platform's terms-of-service, the platform's complaint-resolution process, the platform's recommendation-signal mechanics, the platform's reputation system, the platform's payment-flow timing, and the platform's jurisdictional-coverage rules. Most of these are not substantive commercial decisions; they are infrastructural-property verifications the platform asks the participant to perform on its behalf. The cognitive load is correspondingly higher.

The bonded architecture's interface cognition is structurally lower because the architectural properties are enforced rather than verified. A participant does not need to read the terms-of-service of a permissionless protocol; the protocol's behavior is its terms-of-service, and the protocol cannot violate its terms because the terms are encoded in the bytecode. This shifts the participant's cognitive load from infrastructural verification to substantive commercial decision-making, which is the right direction for a coordination architecture.

The risk of over-loading the interface. A counter-argument is that the bonded architecture's interface may over-load participants who are unfamiliar with cryptographic key management, on-chain transactions, gas estimation, and similar substrate-tier concerns. This is a real concern at the runtime tier and is the work of the wallet, signing, and onboarding infrastructure rather than of the kernel itself. The kernel is incentive-legible; the runtime tier may or may not be, depending on how it presents the kernel to participants. Interface design at the runtime tier is therefore where the architecture's behavioral promises are kept or broken in practice. The kernel makes the promises tractable; the runtime delivers them.

8 Open Empirical Questions

The architecture's behavioral analysis above is theoretical. The experimental questions it invites are several.

Bilateral cooperation rates. The first-order empirical question is whether participants in a bonded bilateral commitment in fact converge on cooperation at rates significantly higher than the conventional contractual baseline. The experimental design is a controlled bilateral exchange where one treatment uses bonded commitment and the control uses conventional contract; the dependent variable is the cooperation rate. We predict significantly higher cooperation under the bonded treatment, with the loss-aversion-amplified kink producing the bulk of the effect.

Multi-party coordination success. The second-order question is whether N-party processes resolve successfully at rates that match the dominance-solvability prediction. The Van Huyck literature establishes that conventional multi-party coordination decays with N; the present paper's argument is that the bonded mechanism does not exhibit this decay because the per-player game is dominance-solvable. An experiment varying N across treatments would test the prediction directly.

Peer-pressure activation. The third-order question is whether the externality structure of atomic resolution is behaviorally activated as peer pressure. The experimental design measures whether participants in an N-party bonded process modify their behavior in response to other participants' visible bond exposure — whether peer monitoring activates at the rates the externality magnitude would predict. The Grameen literature provides the upper-bound benchmark; we expect the bonded mechanism's activation to fall between the Grameen rate and the conventional-contract rate, with the position depending on how much the social-substrate channel contributed to Grameen relative to the joint-liability channel.

Cognitive load measurement. The fourth-order question is the architecture's interface cognition relative to the alternatives. The experimental design measures cognitive load (response time, error rate, self-reported difficulty) across treatments where participants interact with bonded commerce, platform-mediated commerce, and direct off-platform commerce. We predict the bonded treatment's load is intermediate between platform-mediated (highest, due to infrastructural verification) and direct off-platform (lowest, due to absence of architectural superstructure). Whether the architecture's load is significantly lower than platform-mediated in operational deployment is the question that determines the adoption story.

Field measurements. Beyond the laboratory, field measurements of the architecture's operational performance in deployments (the Spirit Air assembly of Paper J, the TradeLens replacement of Paper K, the local-commerce reference assembly of the runtime documentation) will produce the binding evidence on whether the behavioral predictions hold in operational settings. Laboratory results are suggestive; field results are dispositive.

9 Conclusion

The bonded settlement primitive's full-rationality equilibrium analysis carries over to the behavioral setting through four specific properties. The weak-dominance argument operates at the kink between cooperation and defection rather than at the magnitude differences within either domain, where loss aversion sharpens the comparison rather than blunting it. The Van Huyck coordination-failure result does not transfer because the per-player game is dominance-solvable. The atomic-resolution externality structure activates the same peer-pressure dynamics the Grameen literature documents in joint-liability microfinance, through a cleaner architectural channel that does not require the social substrate to be in place. The architecture exhibits incentive legibility: the cognitive operation the mechanism asks of participants is a single comparison at the margin, well within the documented capabilities of unsophisticated participants.

The four properties together produce a mechanism whose behavioral robustness derives from its structural simplicity. The mechanism is not optimal in any technical sense; it is robust in the behavioral sense that it does not ask participants to do cognitive work they cannot reliably do. This is a different property than rational-choice optimality, and it is, we suggest, the more important property for any mechanism that aspires to operate at population scale among real participants.

The contribution of this paper is the behavioral analysis. Combined with the mechanism-design treatment, the argument is that the bonded primitive is both formally elegant under full rationality and behaviorally robust under documented departures from full rationality. Whether the behavioral predictions hold experimentally and in field deployment is the question the architecture invites and that we identify as the experimental work future research should undertake.

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